

ORIGINAL INVESTIGATION

WILEY

Computed tomographic lymphangiography via intra-metatarsal pad injection is feasible in dogs with chylothorax

Lee-Shuan Lin^{1,2}  | Hsien-Chieh Chiu³ | Ryohei Nishimura⁴ | Reina Fujiwara⁵ | Cheng-Shu Chung⁶

¹Laboratory of Veterinary Diagnostic Imaging, Department of Veterinary Medicine, College of Veterinary Medicine, National Pingtung University of Science and Technology, Pingtung, Taiwan

²UniCore Animal Hospital, Taipei City, Taiwan

³Tzuoo Ann Animal Hospital, New Taipei City, Taiwan

⁴Laboratory of Veterinary Surgery, Graduate School of Agriculture and Life Sciences, The University of Tokyo, Japan

⁵Department of Diagnostic Imaging, Veterinary Medical Center, The University of Tokyo, Japan

⁶Laboratory of Veterinary Surgery, Department of Veterinary Medicine, College of Veterinary Medicine, National Pingtung University of Science and Technology, Pingtung, Taiwan

Correspondence

Lee-Shuan Lin, Laboratory of Veterinary Diagnostic Imaging, Department of Veterinary Medicine, College of Veterinary Medicine, National Pingtung University of Science and Technology, No. 1, Shifu Road, Neipu, Pingtung 91201, Taiwan.
Email: linleeshuan@gmail.com

Abstract

Lymphangiography can be useful for preoperative planning in chylothorax. Conventional ultrasound-guided intranodal injection can be difficult in some cases and is dependent upon operator skill. Alternative methods have been proposed to simplify the procedure, but their feasibility has not been sufficiently evaluated in clinical cases. The primary purpose of this multicenter, retrospective, descriptive study was to assess the feasibility and describe the clinical findings of CT lymphangiography by intrametatarsal pad injection in dogs with naturally occurring chylothorax. Twenty dogs were analyzed, and enhancement of thoracic ducts (TDs) was successful in 18 (90%) dogs within 5–14 min after initiating the injection, while successful enhancement of the lymphatic vessels cranial to the popliteal lymph nodes was seen in all dogs within 5 min after injection. The dose with good success to achieve TD enhancement was 1 mL/kg (concentration 350 mg I/kg). Only two dogs had mild discomfort after recovery from general anesthesia. Computed tomographic lymphangiography by intrametatarsal pad injection is a feasible, easy, and safe procedure, which could provide adequate TD and cisterna chyli enhancement, identify TD number and cisterna chyli location and structure, and contribute to surgical planning.

KEYWORDS

chylothorax, dog, intrapad, lymphangiography

1 | INTRODUCTION

Chylothorax in dogs involves chylous accumulation in the thoracic cavity due to abnormal flow or pressure on the thoracic duct (TD) and/or its branches. Chylothorax can be idiopathic or due to an underlying condition such as heart disease, cranial vena cava thromboembolism, mediastinal neoplasia, congenital abnormalities of the heart and TD,

trauma, infection, and foreign objects. Thoracic duct ligation, either by conventional open surgical intervention or minimally invasive approaches, is commonly combined with pericardiectomy and/or cisterna chyli ablation to improve outcome.^{1–4} Although intraoperative lymphangiography by digital subtraction radiography or directly visualization via methylene blue dye injection has been applied to locate the TD and the cisterna chyli,^{2–6} given their variable structure,

Abbreviations: ICG, indocyanine green; MB, methylene blue; TD, thoracic duct.

EQUATOR NETWORK DISCLOSURE: The authors followed the STROBE reporting guidelines in this report.

PREVIOUS PRESENTATION DISCLOSURE: Preliminary results of this study were presented at the EVDI 2019 Scientific Meeting, Basel, Switzerland, Aug 21–24, 2019.

This is an open access article under the terms of the Creative Commons Attribution License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

© 2020 National Marine Mammal Foundation. *Veterinary Radiology & Ultrasound* published by Wiley Periodicals, Inc. on behalf of American College of Veterinary Radiology

preoperative identification of TD branches and cisterna chyli is considered important for surgical planning and to determine the approach site,⁶ as incomplete ligation might result in persistent chylous effusion.⁵

In the past of veterinary medicine, TDs have been imaged by radiography⁷ or computed tomography (CT)^{8,9} immediately after injection of iodinated contrast agent into the mesenteric lymphatic vessels, intraoperatively. This technique demonstrates the lymphatic vessels in excellent details, but requires an invasive laparotomy procedure.⁷ Percutaneous ultrasound-guided injection of contrast agents into lymph nodes has been proposed as a safe and effective method for CT lymphangiography. In dogs, CT lymphangiography via injection of a popliteal lymph node was successful in opacifying the TD in 73% (8/11) of the cases,⁸ whereas injection of mesenteric lymph nodes was successful in opacifying the TD in 100% of dogs ($n = 6$) with chylothorax.¹⁰ However, the skills of an experienced operator are required in these procedures, which can often be difficult even with several attempts.^{5,8} In particular, it is not easy to inject a large amount of contrast agent into the lymph nodes of a small-breed dog.

CT lymphangiography by subcutaneous injection of the contrast agent into perianal tissue had been proposed as a simpler method,^{11,12} but infection remains a cause of concern.¹³ A recent publication¹⁴ described the use of subcutaneous iohexol injection into the dorsal metatarsal region for CT lymphangiography in a sample of five normal dogs and one dog with chylothorax: The TD was identified when at least 0.75 mL/kg of iohexol was injected and this may be a simpler procedure than ultrasound-guided injection of a lymph node. However, only one clinical case was examined, and the potential needle trauma to the tendons of digits and nerves during injection could not be neglected due to the thin subcutis at the dorsal metatarsal region, especially in small breeds. The footpad, which is made of thick subcutaneous adipose tissue for cushion, is also a convenient injection site, and has been widely used in rodent studies.^{15–18} Through footpad injection, the contrast agents migrate into the lymphatics from the interstitial tissue, allowing visualization of lymphatic vessels and TDs.¹⁹

Here, we retrospectively reviewed CT lymphangiographic images obtained by intrametatarsal pad injection in canine patients with naturally occurring chylothorax to assess the feasibility of intrametatarsal pad injection and the quality of the ensuing CT lymphangiography and to describe the related clinical findings.

2 | MATERIALS AND METHODS

2.1 | Case selection

This multicenter, retrospective, descriptive study included medical and image records gathered from the Tzuoo Ann Animal Hospital (New Taipei City, Taiwan), the UniCore Animal Hospital (Taipei City, Taiwan), and the Veterinary Medical Teaching Hospital of National Pingtung University of Science and Technology (Pingtung, Taiwan), regarding animals presented from November 1, 2017 to August 31, 2019. Use of the data was approved by each hospital. Client-owned dogs with chy-

lothorax that underwent CT lymphangiography by intrametatarsal pad injection with non-ionic iodinated contrast agent were included. Dogs were excluded if they had undergone previous surgery associated with TD ligation because the artificial obstruction of lymphatic flow could affect the evaluation. Signalments, CT images, and follow-up information were obtained from the medical records or by telephone interviews with the clients. All inclusion/exclusion decisions were made by the primary investigator (L.S.L.) who is a veterinarian with radiology expertise and experienced researcher.

2.2 | Data recording

2.2.1 | Clinical data

A second-year veterinary radiology resident (W.Y.S.) reviewed the medical records of each dog and recorded the relevant signalments and clinical data, including sex, breed, body weight, and follow-up. The investigator recorded the following CT imaging parameters: CT system, scanned field of view, slice thickness, kV, mAs, pitch, reconstruction matrix, and algorithm used. For the intrametatarsal pad lymphangiography technique, variables recorded were type of contrast medium, injected volume, injection method, and intervals of delayed scans, as were signs of discomfort or adverse effects during and after lymphangiography, surgical follow-up, and outcome.

2.2.2 | Imaging analysis

Computed tomography images were viewed in the soft tissue window (window level 40, window width 300) using commercial medical imaging software (Osirix MD, ver. 10.0.2, Pixmeo SARL, Switzerland). All images were assessed and the final qualitative and quantitative evaluation was made by consensus between the two observers: a veterinarian with radiology expertise (L.S.L.) and one with surgery expertise (C.S.C.). Both observers could see the patient ID and name on the images but were unaware of the detailed signalment, injected contrast volume, injection method, and outcome at the time of evaluation. The images were postprocessed using an imaging workstation (Attractive, PixSpace Ltd., Fukuoka, Japan).

Identification of lymphatic vessels and TD enhancement

We defined the lymphangiography as successful if the contrast medium could migrate to the lymphatic vessels cranial to popliteal lymph nodes; and successful TD enhancement if the contrast medium successfully highlighted the TD. The time elapsed from injection to migration to the lymphatic vessel cranial to popliteal lymph nodes and to TD enhancement was recorded.

Qualitative and quantitative evaluation of TD enhancement

At each mid-vertebral region, from T9 to L1, the number of visible TD branches was assessed and recorded. At the caudal thorax, at the T10–12 level, the TD branch location relative to the aorta was expressed using a clock-face analogy, as previously described,⁸ with slight modifications. The quality of TD enhancement at the T10–12 level was graded as follows: grade 3 (excellent) = all branches strongly enhanced; grade

2 (good) = strongly enhanced main branch, suboptimal enhancement of smaller branches; grade 1 (Fair) = all branches faintly enhanced and difficult to track; 0 (poor) = no enhancement. If contrast leakage was noted, this was recorded.

Evaluation of the cisterna chyli

Cisterna chyli included in the scanning field were evaluated and their location relative to the aorta was described.

Efferent lymphatic pathway evaluation

The pattern of the efferent lymphatic pathway was also assessed if hindlimbs and pelvic region were included in the scan field.

2.3 | Data analysis

Statistical analysis was performed by a single author (C.S.C.) using a spreadsheet (Microsoft® Excel for Mac, version 16.16.11, Redmond, WA, USA) and commercial software (Prism, ver. 8.0, GraphPad Software Inc., La Jolla, CA, USA). The percentages for categorical data were calculated. The age and body weight were expressed as median and range because a preliminary Shapiro-Wilk test demonstrated the non-normally distributed data.

3 | RESULTS

3.1 | Study population

Twenty dogs were included (13 Shiba Inus, 2 Dachshunds, 2 Golden Retrievers, 1 French Bulldog, 1 Shetland Sheepdog, and 1 mixed breed; median age 5.5 years, range 1–13; median weight 9.7 kg, range 5.9–27.3; 10 spayed females, 5 neutered males, 3 intact males, and 2 intact females).

3.2 | Imaging equipment

Three different MDCT scanners in helical scan mode were used, with slightly different imaging techniques. Technical details and parameters are provided in Supporting Information 1.

3.3 | CT scan and intrametatarsal pad lymphangiography

Animals were anesthetized and positioned in sternal recumbency. Helical CT acquisition was initiated in the craniocaudal direction at the UniCore Animal Hospital and Veterinary Medical Teaching Hospital of National Pingtung University of Science and Technology, and in the caudocranial direction at the Tzuoo Ann Animal Hospital. Computed tomography scans were performed by different veterinarians on duty on the examination day. After the localizing acquisition, helical pre-contrast scanning was performed. The scanning included thoracic and abdominal regions in 18 dogs, but only spanned from the thoracic inlet to mid-body of L2 or L5 level in two dogs.

Lymphangiography was then performed by injection of non-ionic iodinated contrast agents into the bilateral metatarsal pads in equal aliquots by two veterinarians at the same time. Following aseptic preparation, the contrast agent was injected into the adipose tissue at 45° to the surface, at 1-cm depth, using a 21- or 22-gauge needle. The injection was completed within 15–30 s, followed by massage of the metatarsal pads, legs, hocks, and thighs for 3–4 min (Figure 1; Supporting Information Video). A CT scan was performed 5 min after initiating the intrametatarsal pad injection. If the contrast agent failed to opacify the TD within 5 min after lymphangiography, serial delayed scans with 30 s–1 min intervals were performed until satisfactory TD-enhancement was obtained. Thoracic duct enhancement was not achieved after serial scans in two dogs, and scanning was discontinued at 30 and 40 min after the injection.

The median dose of contrast agent injection needed to achieve TD enhancement was 1 mL/kg. As shown in Supporting Information 2, the injection volume was 1 mL/kg in 15 dogs. Three dogs received single injections of 0.65, 1.25, and 1.5 mL/kg, respectively. One dog was injected with 1 mL/kg without achieving TD enhancement initially, and an additional 0.7 mL/kg bolus of contrast agent resulted in better TD enhancement within 5 min after the second injection. Another dog received 0.4 mL/kg contrast injections six times, but no TD enhancement was achieved.

Postcontrast CT scan with intravenous contrast administration was then performed in a helical mode after lymphangiography had been performed. The volume of intravenous contrast agent ranged from 1 to 2 mL/kg.

The contrast agent used at the Tzuoo Ann Animal Hospital and the UniCore Animal Hospital was iohexol (Omnipaque® 350, GE Healthcare, Carrigtwohill, Ireland), while ioversol (Optiray® 350, Guerbet, Quebec, Canada) was used at the Veterinary Medical Teaching Hospital of National Pingtung University of Science and Technology. The concentration was 350 mg I/kg.

3.4 | Timing of lymphatic vessel and TD enhancement

Enhancement of the lymph vessels cranial to the popliteal lymph node was identified within 5 min after injection in all dogs, indicating 100% successful lymphangiography by this method.

Thoracic duct enhancement occurred in 18 dogs (90%) within 14 min after injection (Supporting Information 2): TDs were successfully identified within 5 min in 13 dogs (65%) in the first scan, and within 5–14 min by repeated scanning in four dogs (20%); one dog (5%) received two bolus contrast injections, and TD enhancement was identified within 5 min after injecting the second bolus (21 minutes after the first contrast injection). No TD enhancement was found in 2 dogs (10%): one received a single contrast injection, followed by repeated CT scanning for 30 min (the contrast was identified up to the L3 region); the other received six injections (0.4 mL/kg/injection) at 5-min intervals, followed by repeated CT scanning for a period of 40 min in total (the contrast was identified up to the T13 region).



FIGURE 1 Intra-metatarsal pad computed tomography (CT) lymphangiography. Illustration (A) and photographs (B) of bilateral intra-metatarsal pad injection of contrast agent for CT lymphangiography. The pads (C) and legs (D) are massaged for 3–4 min after the injection [Color figure can be viewed at wileyonlinelibrary.com]

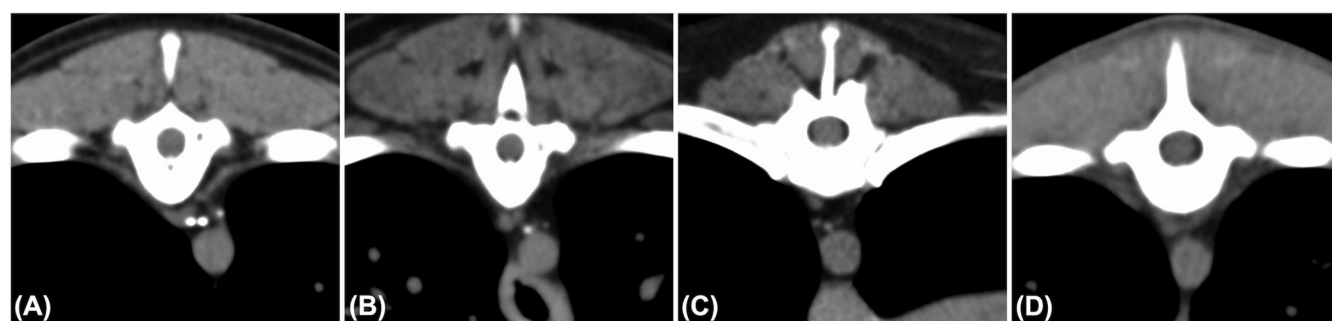


FIGURE 2 Grading and percentage of thoracic duct (TD) enhancement at the T10–12 level. A, Grade 3 (excellent) = all branches are intensively enhanced, 65%; B, grade 2 (good) = the main branch is intensively enhanced with suboptimal enhancement of smaller branches, 10%; C, grade 1 (fair) = all branches are faintly enhanced and difficult to track, 15%; D, grade 0 (poor) = absence of enhancement, 10% (transverse CT images, 40 window level, 300 window width, soft tissue algorithm, 1 mm slice thickness)

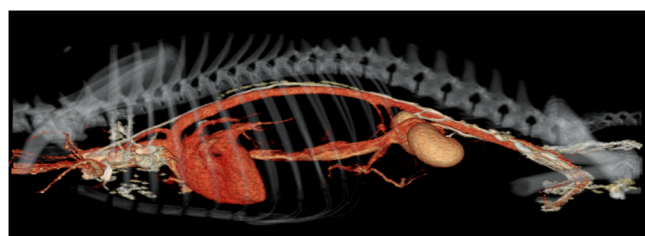


FIGURE 3 Representative 3D image of a dog with chylothorax. The image is post-processed via fusion of lymphangiography and CT angiography. The whole lymph vessels from pelvic region to the TD branches are well visualized, and the anatomic position between vasculature and lymph vessels could also be identified. Contrast leakage is recognized at the level of the cranial mediastinum [Color figure can be viewed at wileyonlinelibrary.com]

3.5 | Quality and quantity of TD enhancement

The quality of TD enhancement at the T10–12 level was as follows (Figure 2): Thirteen cases (65%) exhibited grade 3 enhancement (excellent), two (10%) exhibited grade 2 enhancement (good), three (15%) exhibited grade 1 enhancement (fair), and two (10%) exhibited grade 0 enhancement (poor). Figure 3 illustrates a representative 3D image, post-processed via fusion of lymphangiography and CT angiography, of a dog with grade 3 enhancement. The lymphatic vessels and TDs,

TABLE 1 Thoracic ducts (TDs) identified by computed tomography lymphangiography ventral to each mid-vertebral body in the caudal thorax of 18 dogs in which TD enhancement was achieved. The number of dogs is reported according to the vertebral location and TD branches

TD branches	Vertebral location					
	T9	T10	T11	T12	T13	L1
1	10	11	13	12	10	13
2	4	4	5	4	7	5
3	4	2	0	2	1	0
4	0	1	0	0	0	0

T9–T13, thoracic vertebra 9–13; L1, lumbar vertebra 1.

as well as the relative anatomic position of the TDs to the vasculature, are well visualized.

Among 18 dogs with successful TD enhancement, multiple TD branches were identified in 13 dogs (72%). The maximum number of TD branches was as follows: a single branch in five dogs (28%), two branches in eight dogs (44%), three branches in four dogs (22%), and four branches in one dog (6%). The number of TD branches at different vertebral segments varied widely, and is displayed in Table 1.

The clock-face analogy of the TD branches within the T10–12 vertebral segment is shown as a bar-graph in Figure 4: 78% of TD branches were in the right hemithorax, between the 9 o'clock and 12

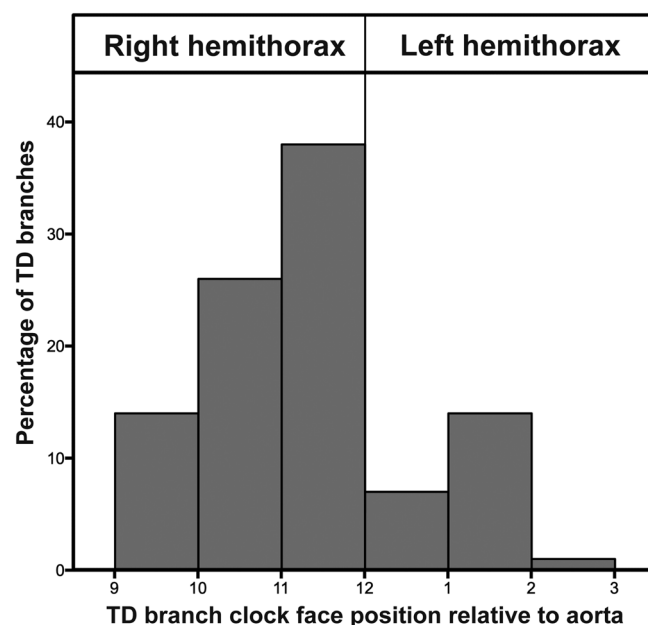


FIGURE 4 Thoracic duct branch location with respect to the aorta for the T10–12 vertebral segment using a clock-face analogy

o'clock positions relative to the aorta, from a craniocaudal view, while 22% were in the left hemithorax between the 12 o'clock and 3 o'clock positions. Contrast leakage to the mediastinum was identified in three dogs, and to the pleural cavity in two dogs.

3.6 | Cisterna chyli location and structure

Scanning fields that included the cisterna chyli were acquired in 18 dogs with successful lymphatic vessel enhancement. The location of the cisterna chyli ranged from L2 to L5, the most common site being L3 (10/18; see Supporting Information 2). Three of these dogs did not have an apparent cisterna chyli structure. Figure 5 illustrates the six different types of cisterna chyli structure identified: left to the aorta (1/18); right and dorsal (7/18); ventral, right and dorsal (3/18); right, dorsal and left (1/18); dorsal, left and ventral (1/18); or circling the aorta (2/18).

3.7 | Lymphatic pathway assessment

The scanning field included the hindlimb and pelvic regions in 18 dogs, in which the lymphatic vessels directly drained from the popliteal lymph nodes to the medial iliac lymph nodes along the proximal pelvic limb. Three dogs (17%) exhibited bilateral accessory popliteal efferent lymphatic vessels, coursing to the internal iliac and/or sacral lymph nodes from the dorsal gluteal region, while three (17%) dogs had only unilateral accessory efferent lymphatic pathways (Figure 6).

3.8 | Follow-up

Six dogs had been diagnosed with concurrent disease and considered as non-idiopathic chylothorax cases: tricuspid regurgitation ($n = 2$), right middle lung lobe necrosis ($n = 1$), tension pneumothorax ($n = 1$),

bronchopneumonia ($n = 1$), and thrombosis ($n = 1$) at the left brachiocephalic vein and right renal vein.

During intrapad injection, increased heart rate was briefly observed, which gradually returned to normal in all dogs after pad and leg massage. After recovery from anesthesia, only two dogs had mild signs of discomfort: one dog showed swelling of the paws for one night, with contrast leakage, while another dog, which showed swelling of the hindlimbs for 1 week, had undergone intrapad injection for CT lymphangiography, followed by intrapad injection of indocyanine green (ICG) and methylene blue (MB) for intraoperative lymphangiography on the same day.

Fifteen dogs underwent video-assisted thoracoscopic surgery: TD ligation with pericardiectomy alone in 13 dogs, and TD ligation with pericardiectomy combined with cisterna chyli ablation via laparoscopy in two dogs. The success rate of the first surgery was 93% (14/15) regardless of the surgical procedure. Five dogs in a more recent cohort also underwent intra-operative ICG/Near-infrared fluorescence (NIRF) lymphangiography and 10 dogs had direct visualization of the TDs intraoperatively. Approach sites were chosen according to the CT lymphangiographic findings, concordant intraoperative identification of the TD branches was reported. However, one dog still had persistent chylothorax, and postoperative CT lymphangiography revealed an additional branch that had not been identified in preoperative lymphangiography. The dog underwent a second surgery in combination with intraoperative ICG/NIRF lymphangiography via intra-metatarsal pad injection to ensure complete ligation, and recovered well. Preoperative CT lymphangiography was not successful in one dog, but intraoperative ICG/NIRF lymphangiography was achieved 15 min after ICG injection into metatarsal pads. One dog recovered well from the surgery but was euthanized for cardiac neoplasia after 3 months. Among the five dogs who did not have surgery, one underwent thoracic tube placement to remove the chyle with concurrent medical treatment for 5 days and responded well; two did not receive any treatment, including one that died 3 months later and one lost to follow-up; the dog with tension pneumothorax mentioned above died the night after the CT examination; the dog with concurrent thrombosis died 5 days after the CT examination, before the scheduled surgery.

4 | DISCUSSION

We report a CT lymphangiography technique using intra-metatarsal pad injection in dogs as a feasible, relatively easy procedure with high success rate (90% in the current study), providing adequate identification of lymphatic vessels, cisterna structures, and TD branches in chylothorax cases. We recommend the following procedure: Prepare at least 1 mL/kg contrast medium, divide it into two equal aliquots, and inject into the metatarsal pads bilaterally; thoroughly massage the pads and legs for 3–4 min, then start the first CT scan within 5 min after initiating the injection. If the TDs are not well opacified, perform serial delayed scans at 30–60 s intervals until satisfactory TD enhancement is achieved.

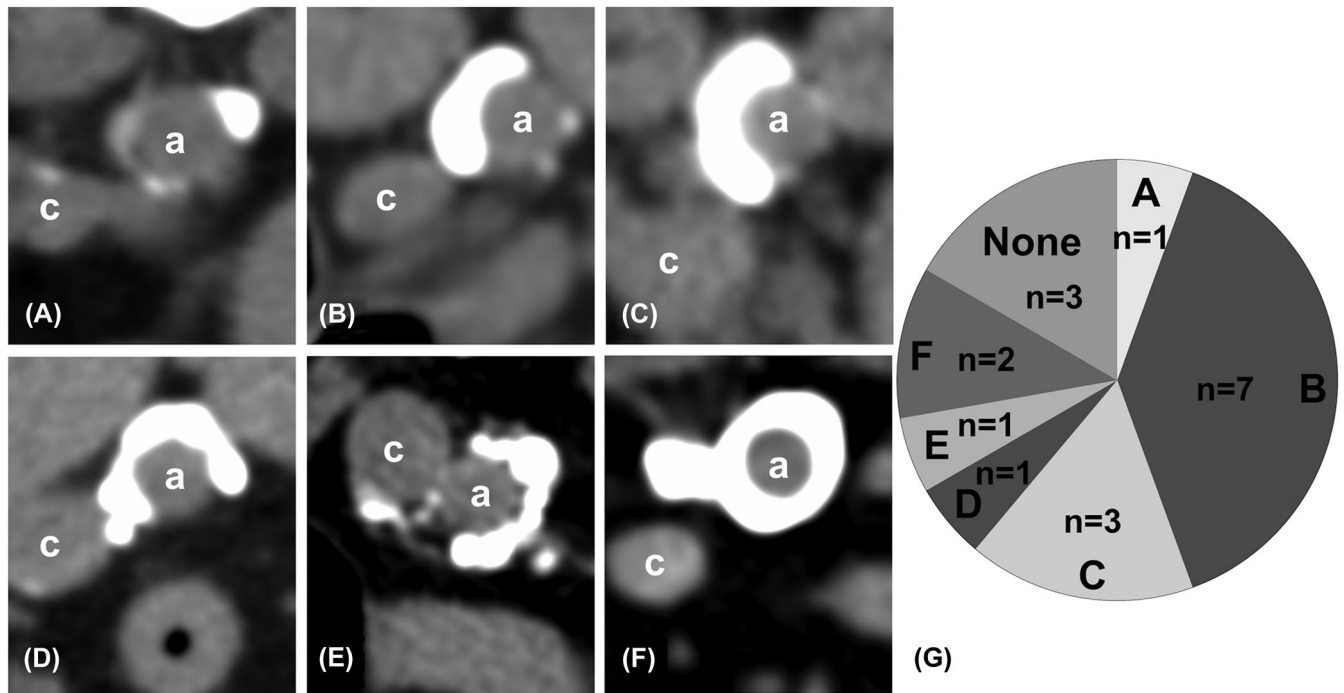


FIGURE 5 Highly variable cisterna chyli structure among individuals. Different structural types of the cisterna chyli are seen with respect to the aorta: left to the aorta (A); right and dorsal (B); ventral, right and dorsal (C); right, dorsal and left (D); dorsal, left and ventral (E); or circling the aorta (F). G, prevalence of each type among the 18 dogs with enhanced cisterna chyli. a, aorta; c, cauda vena cava; none, no apparent cisterna chyli structure is identified (transverse CT images, 40 window level, 300 window width, soft tissue algorithm, 1 mm slice thickness)

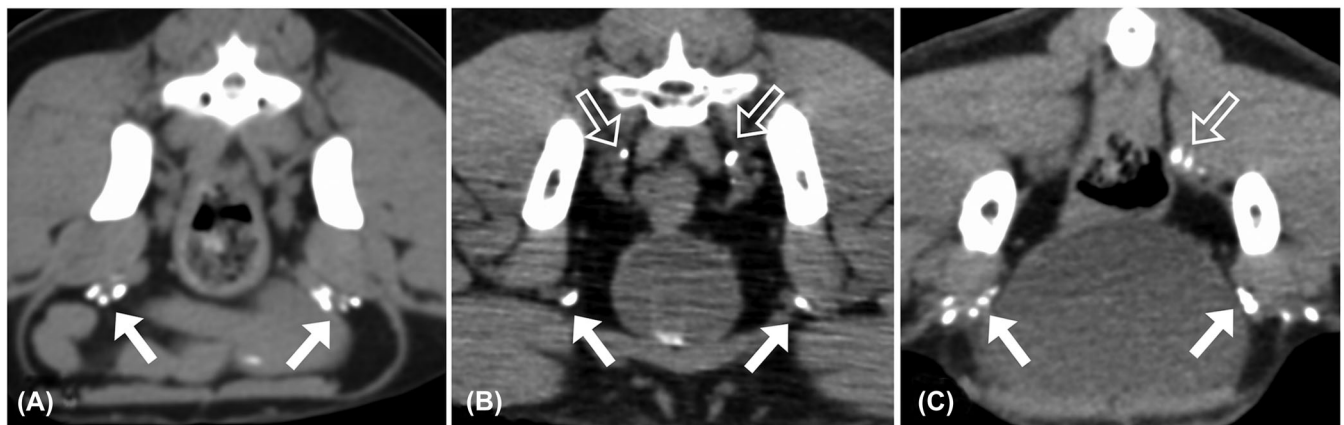


FIGURE 6 Efferent lymphatic pathways of the popliteal lymph node. The white arrows indicate lymphatic vessels directly draining from the popliteal lymph nodes to the medial iliac lymph nodes, along the medial thigh, while the open arrows indicate accessory popliteal efferent pathways from the dorsal gluteal regions. The dogs exhibited a bilateral medial pathway (A), bilateral medial and accessory pathway (B), or unilateral accessory pathway (C). (transverse CT images, 40 window level, 300 window width, soft tissue algorithm, 1 mm slice thickness)

Ultrasound-guided injection of contrast into the popliteal and mesenteric lymph nodes for CT lymphangiography has been proposed as an acceptable alternative procedure to a surgical approach for mesenteric lymphangiography and achieved similar visualization of the TD branches. However, a lower success rate (73%) was reported⁸ when attempting injection into the popliteal rather than into the mesenteric lymph nodes (100%).¹⁰ Another study⁵ reported an 86% success rate of CT lymphangiography by injection of iopamidol into mesenteric lymph nodes and subsequently medial iliac or popliteal lymph nodes

in cases in which the mesenteric lymph nodes was unsuccessful. These studies all mentioned the difficulties of injection into the lymph nodes, which are sometimes small and unable to accept a reasonable volume of contrast, even after several attempts. A recent publication¹⁴ evaluating subcutaneous iohexol injection into the dorsal metatarsal region in five healthy dogs and one chylothorax case reported a 100% success rate. In the current study, we collected 20 clinical chylothorax cases, and a high success rate of TD enhancement (90%) was demonstrated, with excellent (grade 3) or good (grade 2) quality enhancement

achieved in 75% of the dogs. Intrametatarsal pad injection provides an alternative, simple method for CT lymphangiography and exhibits success rates in chylothorax cases comparable to those of commonly used techniques.

Identification of the TD branches is important for successful TD ligation⁵ because a continuous flow of chyle may persist if even one TD branch is not occluded.²⁰ In our study population, 78% of the TD branches at T10–12 level were located to the right of the aorta, and in some cases the number of TD branches was highly variable over a short distance, in agreement with a previous study.⁸ Surgeons decided the approach site according to the CT lymphangiographic findings, and tried to ligate the TDs at the T10–11 level, where the branches were typically 1 or 2, making complete ligation easier to achieve. TD ligation combined with pericardiectomy in all cases and cisterna chyli ablation in some cases led to resolution of chylothorax in most dogs (93%) at first surgery. Although the surgeons reported concordant intraoperative identification of TD branches, persistent chylothorax still occurred in one dog. Postoperative CT lymphangiography revealed an additional branch not seen in the preoperative images. The TD enhancement quality in this dog was classified as “excellent,” suggesting that TD branches might be underestimated even with excellent enhancement. However, the application of intraoperative NIRF lymphangiography via intrametatarsal pad injection of ICG could aid in real-time identification of the TD branches or small branches not seen in preoperative CT lymphangiography, and ensure complete ligation.

Cisterna chyli ablation with TD ligation has been proven to improve the outcome of chylothorax³; therefore, correct preoperative localization of the cisterna chyli is important in surgical planning.⁵ In our study, 16% of the dogs did not exhibit a clear regional lymphatic dilatation identifiable as a cisterna chyli structure, a fraction slightly lower than previously reported (23%).⁵ Hayashi et al and Sakals et al performed a left-sided approach to cisterna chyli ablation, because the right-side approach might increase the risk of accidental trauma to the caudal vena cava.^{3,21} Further, Sakals et al described the difficulties in identifying the cisterna chyli structure intraoperatively, resulting in failed ablation in some cases. Indeed, in our study, only 28% (5/18, see Figure 5, types A, D, E, and F) of the cases had an apparent left-side cisterna chyli structure, while 17% (3/18) had no apparent structures. This may explain the occasional failures when all cases were approached from the left-side. A more recent study also emphasized the importance of preoperative lymphangiography for surgical planning, the selection of the side of the abdomen to approach, and the best corridor for cisterna chyli ablation in minimally invasive chylothorax surgery.⁶ In the current study, only two cases underwent cisterna chyli ablation, and the surgeons decided the approach site according to the preoperative lymphangiography; the chylothorax was resolved successfully in these cases. In one of these cases intraoperative ICG/NIRF lymphangiography was used, and successful ablation could be confirmed by visualizing the leakage of the fluorescence. Despite the small number of cases with cisterna chyli ablation in our study, preoperative identification of the location of the cisterna chyli via CT lymphangiography should be considered for surgical planning due to its varied location around the aorta.

Concerning optimal TD enhancing timing, Naganobu et al injected contrast agent into the popliteal lymph node for 5 min and reported the optimal timing for enhancement to be within 2 min after injection.²² Kim et al injected the contrast agent into the dorsal metatarsal subcutaneous region for 2 min, followed by gentle massage of the injection site for 1 min, and observed well-enhanced lymphatic vessels 3, 5, and 7 min after injection.¹⁴ In our study, the intrapad injection time was 15–30 s, followed by a thorough massage of the pads and legs for 3 min, and TD enhancement was successful in 90% of the dogs within 5–14 min after injection. Compared with previous reports, the enhancement timing was similar to that of metatarsal region injection, but slower than injection into the popliteal lymph nodes. Our injection period is shorter than other procedures, possibly due to the larger capacity of the metatarsal pad, and the massage of the pads and legs allowing better migration of the contrast to the lymphatic vessels, as previously described in both human^{23,24} and veterinary¹⁴ literature.

The volume of contrast agent injected can also influence TD enhancement.⁵ Previous studies reported 1 mL/kg contrast volume for popliteal lymph node injection,⁸ and 0.75–1 mL/kg for metatarsal subcutaneous injection in dogs.¹⁴ The contrast volume used in this study was most frequently 1 mL/kg (80%, 16/20) and TD enhancement only failed in one case. Two cases received less than 1 mL/kg contrast injection (0.4, 0.65 mL/kg) and exhibited absent or poor TD enhancement. The other two cases received 1.25 and 1.5 mL/kg contrast injection and no TD enhancement was found in one case. We may conclude that 1 mL/kg is appropriate for intrametatarsal pad injection, while lower volumes might impair the results, and the value of higher volumes requires further investigation.

A previous study described how additional contrast administration may improve TD enhancement,²² and we obtained a similar result in one dog, which received 1 mL/kg contrast agent followed by 0.7 mL/kg. However, TD enhancement failed in one dog, which received six injections of 0.4 mL/kg within 40 min; this was a large-breed dog (golden retriever), and had concurrent tension pneumothorax. Another dog, which received 0.65 mL/kg, was a 27.3-kg mixed-breed dog with concurrent thromboembolism in the left brachiocephalic vein/right renal vein, and exhibited grade 1 (fair) TD enhancement. In these two dogs, the relatively low loading dose and lymphatic occlusion secondary to increased venous pressure may have resulted in poor enhancement.²⁵ Moreover, a contrast agent volume greater than 1 mL/kg may be required for large-breed dogs (>20 kg).

A previous report²⁶ demonstrated accessory popliteal efferent lymphatic pathways in dogs by means of popliteal lymphangiography; this was also achieved in our study by intrametatarsal pad lymphangiography. In our study, 17% of the dogs exhibited bilateral accessory pathway, while 17% exhibited a unilateral pathway. The same report²⁶ also demonstrated communication between bilateral popliteal efferent lymphatic ways, but the contralateral lymphatic flow could not be evaluated in our study because we injected the contrast agents into the bilateral metatarsal pads at the same time.

Although increased heart rate after contrast injection may indicate pain, it soon returned to normal levels after pad and leg massage.

Such pain has been reported as tolerable in a human study of subcutaneous gadoterate meglumine injection into the dorsum of the foot at different interdigital webs for magnetic resonance lymphography.²⁴ Considering the short duration of the stimulation during injection and the normal gait after recovery, additional analgesia for intrapad injection may not be required. For more precise evaluation of the pain or the possible contrast-associated inflammation, an experimental study might be warranted.

Only two dogs showed discomfort or lameness after recovery from anesthesia. Swelling of the paws and contrast agent leakage from the puncture wound, lasting one night, were observed in one dog; while hindlimb edema lasting 1 week was observed in another dog, which underwent further multiple intrapad injections on the same day for subsequent intraoperative lymphangiography. These injections included iodine contrast agent for CT lymphangiography, and ICG and MB. Thoracic duct enhancement in dog #15 was also not successful, the contrast only flowing cranially to the L3 level. Interestingly, unusually slow but successful TD enhancement was achieved in intraoperative ICG/NIRF lymphangiography, possibly due to superior visualization of fluorescence.⁵ A larger study population might be necessary to verify the possible complications of intrapad injection for CT lymphangiography.

This study has some limitations. First, its retrospective nature and the limits of the available data: Not every dog received standardized evaluation for chylothorax, such as an echocardiographic examination. Medical records were insufficient in some cases. Second, multiple institutions were recruited to increase the number of cases, resulting in many different operators having performed intrapad injections and CT exams, which may have affected the results. Third, the majority of the study population were Shiba Inu, with a relative lack of large and giant breeds; whether the recommended procedures are appropriate for such breeds is unknown.

In conclusion, CT lymphangiography by intrametatarsal pad injection is a feasible, easy, and safe procedure, leading to adequate quality of TD and cisterna chyli enhancement, identification of TD number, and cisterna chyli location/structure, which can help in surgical planning. Future prospective studies including a larger sample size and more breeds are needed to establish the proper dosage and the timing of enhancement.

LIST OF AUTHOR CONTRIBUTIONS

Category 1

- (a) Conception and Design: Lin, Nishimura, Fujiwara
- (b) Acquisition of Data: Lin, Chiu
- (c) Analysis and Interpretation of Data: Lin, Chung

Category 2

- (a) Drafting the Article: Lin, Chung

- (b) Revising Article for Intellectual Content: Lin, Chung, Chiu, Fujiwara, Nishimura

Category 3

- (a) Final Approval of the Completed Article: Lin, Chung, Chiu, Fujiwara, Nishimura

ACKNOWLEDGMENTS

We thank the staff of the Tzuoo Ann Animal Hospital, the UniCore Animal Hospital, and the Diagnostic Image Division of the National Pingtung University of Science and Technology, Veterinary Teaching Hospital, for their technical support. We thank Dr. Wan-Ying Shen for helping with data acquisition.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

ORCID

Lee-Shuan Lin  <https://orcid.org/0000-0003-1590-2253>

REFERENCES

1. Fossum TW. *Small Animal Surgery*. St. Louis, MO: Elsevier; 2013. 941 p. Chapter 30, Surgery of the Lower Respiratory System: Pleural Cavity and Diaphragm.
2. Fossum TW, Mertens MM, Miller MW, et al. Thoracic duct ligation and pericardectomy for treatment of idiopathic chylothorax. *J Vet Intern Med*. 2004;18:307-310.
3. Hayashi K, Sicard G, Gellasch K, Frank JD, Hardie RJ, McAnulty JF. Cisterna chyli ablation with thoracic duct ligation for chylothorax: Results in eight dogs. *Vet Surg*. 2005;34:519-523.
4. Mayhew PD, Culp WT, Mayhew KN, Morgan OD. Minimally invasive treatment of idiopathic chylothorax in dogs by thoracoscopic thoracic duct ligation and subphrenic pericardectomy: 6 cases (2007-2010). *J Am Vet Med Assoc*. 2012;241:904-909.
5. Steffey MA, Mayhew PD. Use of direct near-infrared fluorescent lymphography for thoracoscopic thoracic duct identification in 15 dogs with chylothorax. *Vet Surg*. 2018;47:267-276.
6. Morris KP, Singh A, Holt DE, et al. Hybrid single-port laparoscopic cisterna chyli ablation for the adjunct treatment of chylothorax disease in dogs. *Vet Surg*. 2019;48:O121-O129.
7. Birchard SJ, Fossum TW. Chylothorax in the dog and cat. *Vet Clin North Am Small Anim Pract*. 1987;17:271-283.
8. Millward IR, Kirberger RM, Thompson PN. Comparative popliteal and mesenteric computed tomography lymphangiography of the canine thoracic duct. *Vet Radiol Ultrasound*. 2011;52:295-301.
9. Esterline ML, Radlinsky MG, Biller DS, Mason DE, Roush JK, Cash WC. Comparison of radiographic and computed tomography lymphangiography for identification of the canine thoracic duct. *Vet Radiol Ultrasound*. 2005;46:391-395.
10. Johnson EG, Wisner ER, Kyles A, Koehler C, Marks SL. Computed tomographic lymphography of the thoracic duct by mesenteric lymph node injection. *Vet Surg*. 2009;38:361-367.
11. Ando K, Kamijyou K, Hatinode K, Shibata S, Shida T, Asari M. Computed tomography and radiographic lymphography of the thoracic duct by subcutaneous or submucosal injection. *J Vet Med Sci*. 2012;74:135-140.

12. Iwanaga T, Tokunaga S, Momoi Y. Thoracic duct lymphography by subcutaneous contrast agent injection in a dog with chylothorax. *Open Vet J*. 2016;6:238-241.
13. Kim K, Choen S, Hwang J, Jang M, Yoon J, Choi M. CT lymphangiography with contrast medium injection into the perianal subcutaneous region in a dog with chylothorax. *J Vet Clin*. 2018;35:299-301.
14. Kim K, Cheon S, Kang K, et al. Computed tomographic lymphangiography of the thoracic duct by subcutaneous iohexol injection into the metatarsal region. *Vet Surg*. 2019.
15. Yamaji Y, Akita S, Akita H, et al. Development of a mouse model for the visual and quantitative assessment of lymphatic trafficking and function by in vivo imaging. *Sci Rep*. 2018;8:5921.
16. Zhang F, Niu G, Lu G, Chen X. Preclinical lymphatic imaging. *Mol Imaging Biol*. 2011;13:599-612.
17. Hoshi H, Kamiya K, Endo E. Cortical structure of the lymph node. I. Effect of blockage of the afferent lymph flow to mouse popliteal nodes for protracted periods. *J Anat*. 1981;133:593-606.
18. Long KM, Heise M. Safe and effective mouse footpad inoculation. *Methods Mol Biol*. 2013;1031:97-100.
19. Itkin M, Nadolski GJ. Modern techniques of lymphangiography and interventions: current status and future development. *Cardiovasc Intervent Radiol*. 2018;41:366-376.
20. Birchard SJ, Smeak DD, Fossum TW. Results of thoracic duct ligation in dogs with chylothorax. *J Am Vet Med Assoc*. 1988;193:68-71.
21. Sakals S, Schmiedt CW, Radlinsky MG. Comparison and description of transdiaphragmatic and abdominal minimally invasive cisterna chyli ablation in dogs. *Vet Surg*. 2011;40:795-801.
22. Naganobu K, Ohigashi Y, Akiyoshi T, Hagio M, Miyamoto T, Yamaguchi R. Lymphography of the thoracic duct by percutaneous injection of iohexol into the popliteal lymph node of dogs: Experimental study and clinical application. *Vet Surg*. 2006;35:377-381.
23. Lu Q, Delproposto Z, Hu A, et al. MR lymphography of lymphatic vessels in lower extremity with gynecologic oncology-related lymphedema. *PLoS One*. 2012;7:e50319.
24. Ruehm SG, Schroeder T, Debatin JF. Interstitial MR lymphography with gadoterate meglumine: Initial experience in humans. *Radiology*. 2001;220:816-821.
25. Singh A, Brisson BA. Chylothorax associated with thrombosis of the cranial vena cava. *Can Vet J*. 2010;51:847-852.
26. Mayer MN, Sweet KA, Patsikas MN, Sukut SL, Waldner CL. Frequency of an accessory popliteal efferent lymphatic pathway in dogs. *Vet Radiol Ultrasound*. 2018;59:365-373.

SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article.

How to cite this article: Lin L-S, Chiu H-C, Nishimura R, Fujiwara R, Chung C-S. Computed tomographic lymphangiography via intra-metatarsal pad injection is feasible in dogs with chylothorax. *Vet Radiol Ultrasound*. 2020;61:435-443. <https://doi.org/10.1111/vru.12865>